

MATSim – Simulating Hong Kong Transportation Activities



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Background

Hong Kong

- limited road infrastructure
- high travel demand

MATSim

- simulate Hong Kong's urban transport system

What is MATSim?

Definition:

- Multi-Agent Transport Simulation
- Agent-based modeling framework

Purpose:

- Models individual travel behavior (24-hour)
- Evaluates the impact of infrastructure changes, and urban development on traffic flow and agent behavior

What is MATSim?

Framework: three stages

Input:

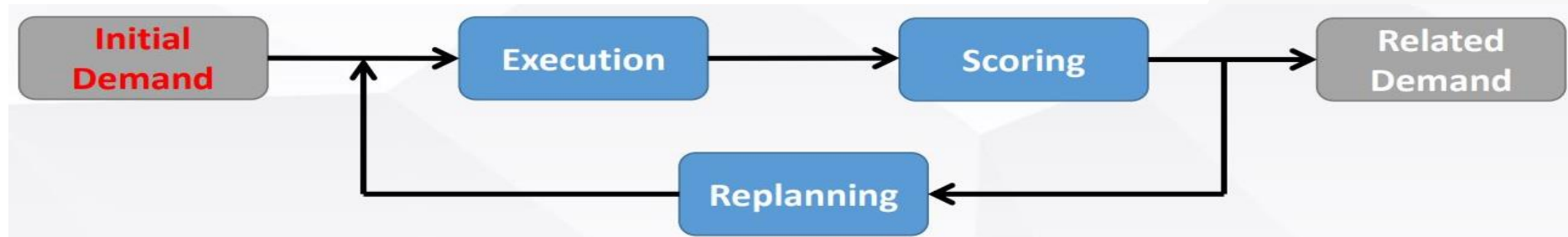
- Defines agents' daily activity schedules, transport networks, and facilities

Simulation Process:

- Execution, scoring, and replanning stages

Output:

- Link statistics, traffic flow data, and agent scores



Rationale – Simulating Hong Kong transport activities

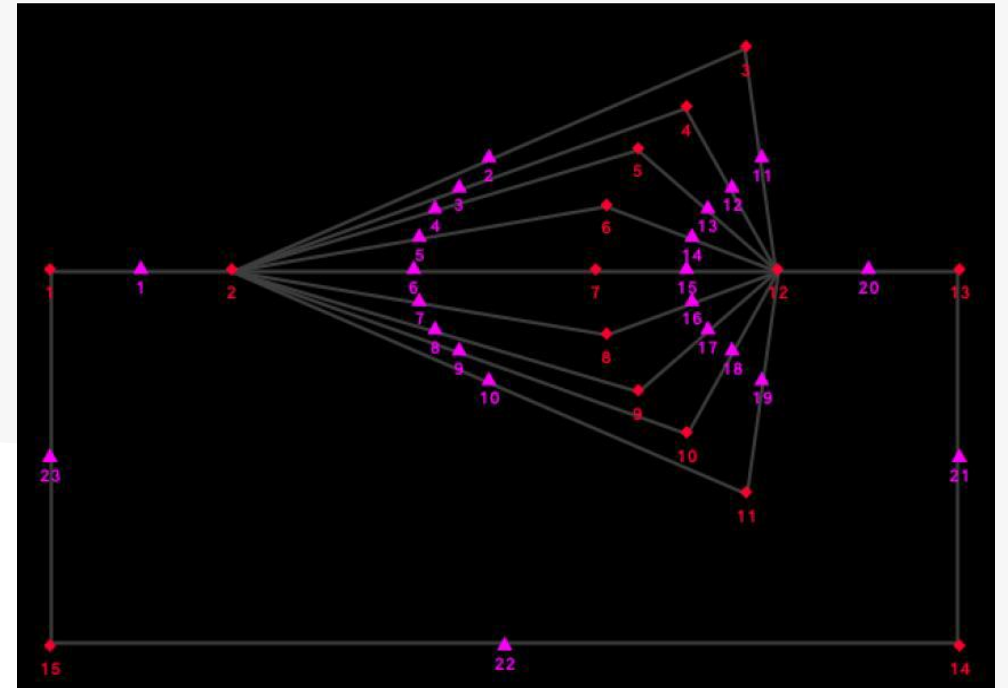
- Represent transport activities in Hong Kong
- Explore the use of MATSim to simulate Hong Kong transport conditions
- A real-world reference to our MATSim model
- Understand how the modifications influence simulation outcomes

Aims of our MATSim scenario

- Simulate the travel agents in Hong Kong
- Data integrity
- Based on real-world data
- How changing lengths, capacities, and freespeed affects the outcome

Our base case - "equil"

- Simple network
- Consists of 15 nodes and 23 facilities
- 100 agents
- Start and end at the same point
- Identical travel purposes



Base MATSim Scenario

Our base case - "equil"

Simple network

- controlled environment
- isolate variables
- clear visibility of cause-and-effect relationships
- trace the impact of each change

Customizations - Overview

1. Generating 500 agents
2. Two-way roads
3. Freespeed
4. 3 Scenarios, each with different lengths and capacities
5. Generate with 2 different number of iterations

Customization 1 – Generating 500 agents

Overview

- 500 agents – max agents with Via
- Generated using a Python script
- Generate daily plans (plans500.xml)
- Activities: home-based work/school/other
- Graph plotter in Python – visualization

Generating 500 agents (cont.)

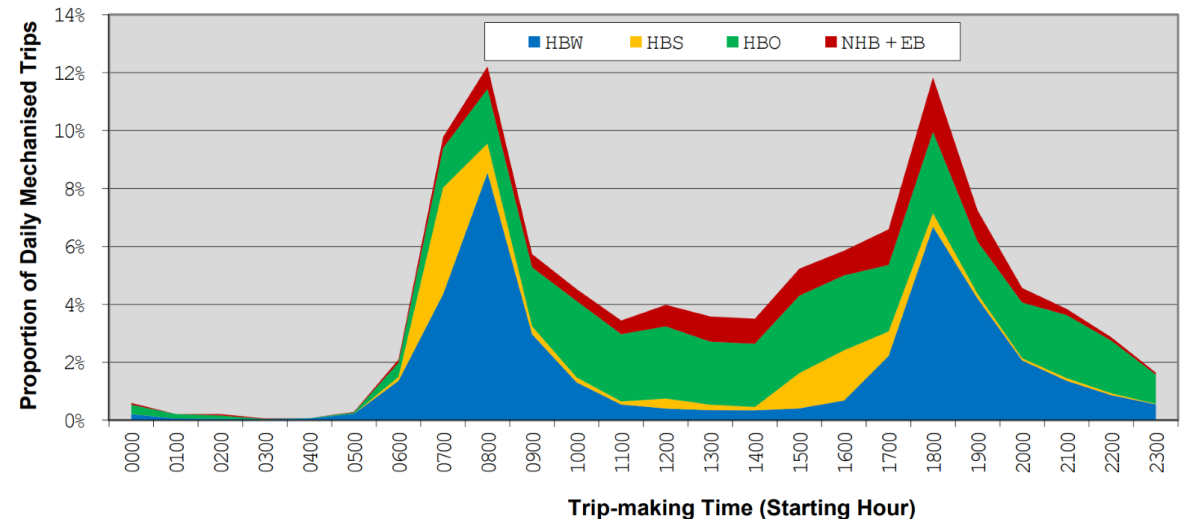
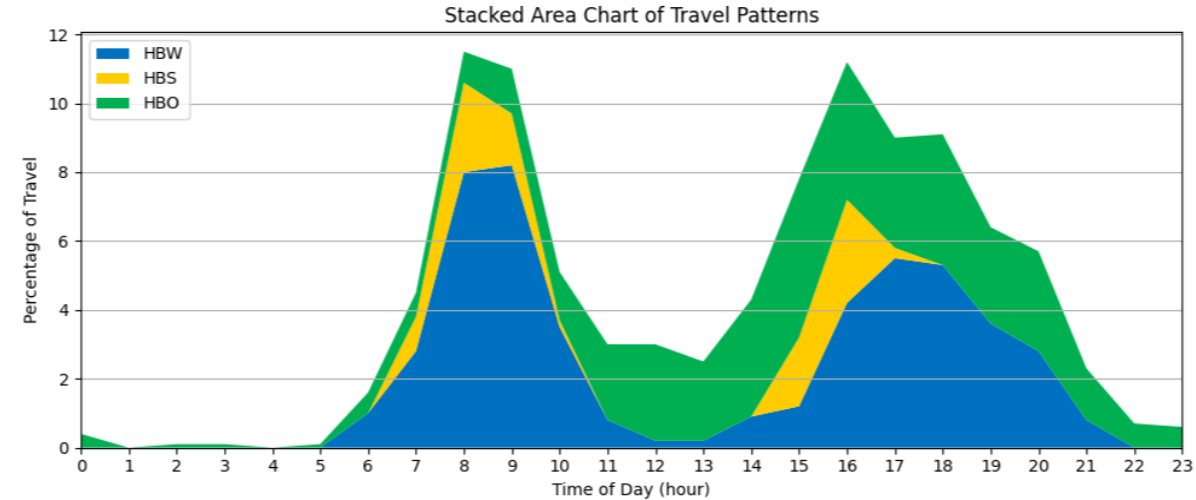
- Based on the survey from 2011 Transport Department
- Minimalized to 500 agents
- Start on a random link
- End on a random facility
- Random purpose of travel (work, school, other)

Rationale behind generating 500 agents – Data integrity

- our aim: to simulate a minimized version of real-world data
 - Travel data is from the Survey by Transport Department
 - Accuracy is our major concern
- 

Rationale behind generating 500 agents – Data integrity (cont.)

- **Representativity and Limitations**
- Overlapping to some extent
- No NHB + EB (red)
- NHB is non-home to non-home
- EB is work to work
- Excluded: extra layer of complexity



Customization 2 – Two-way roads

- Roads in Hong Kong are bidirectional
- Extra links with reversed nodes
- Road links increased from 23 to 46
- Same road nodes
- Different link id

Customization 3 - Freespeed

- 50 km/h - general speed limit in Hong Kong
- 50 km/h is 13.88 m/s
- Freespeed is set to 14.88, +1 m/s
- Reference to real-world data

Customization 4 – Different Capacities

- The maximum number of vehicles that can pass a link per hour
 - Adjusted to reflect different road types and traffic conditions
- 
- A decorative graphic at the bottom of the slide consisting of a light gray wavy line. On the right side of this line, there are two white circles of different sizes, one slightly overlapping the other.

Customization 4 – Different Capacities

Scenario 1

- **Capacities:** 250 and 2500 vehicles/hour.
- from TD statistics: approximately 77 private cars owned per 1,000 people
- our agent number: 500 → about 40 private cars
- so, the capacity of 250 is more than enough for the simulation network
- 250 is suitable for larger agent populations

Customization 4 – Different Capacities

Scenario 2

- **Capacities:** 200 and 100 vehicles/hour.
 - Tests whether slight capacity changed can reduce congestion compared to Scenario 3
- 
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Customization 4 – Different Capacities

Scenario 3

- **Capacities:** 80 and 40 vehicles/hour.
- Mimics real-world congestion
- Agent number: 500 → 40 private cars
- Reflect more congested urban roads

Customization 5 – Different Length

- the physical road distance of a link in meters
- Adjust to reflect dense urban areas and test the impact on congestion condition

Customization 5 – Different Length

- Base "equil" scenario: 2000m, 5000 m, 10000m and 35000 m
- Reference figures from TD statistics: length of public roads per 1000 population is approximately 298 m
- Scenario 1: 5000m, 10000m; Scenarios 2, 3: 750m, 1500m
- together with decreased road capacity
- Reasons:
 - (a) scenarios 2, 3 parameters changed close to real-world data, represent short-length roads and highly dense areas that commonly found in HK
 - (b) to find out how the average trip duration and travel speed would change

Customization 6 – Different number of Iterations

- **Iterations:** 20, 50 for scenarios 1, 2, 3
- Better route optimization with capacity constraints
- Reason: to find whether the score of daily plan, average travel time would change with greater iterations

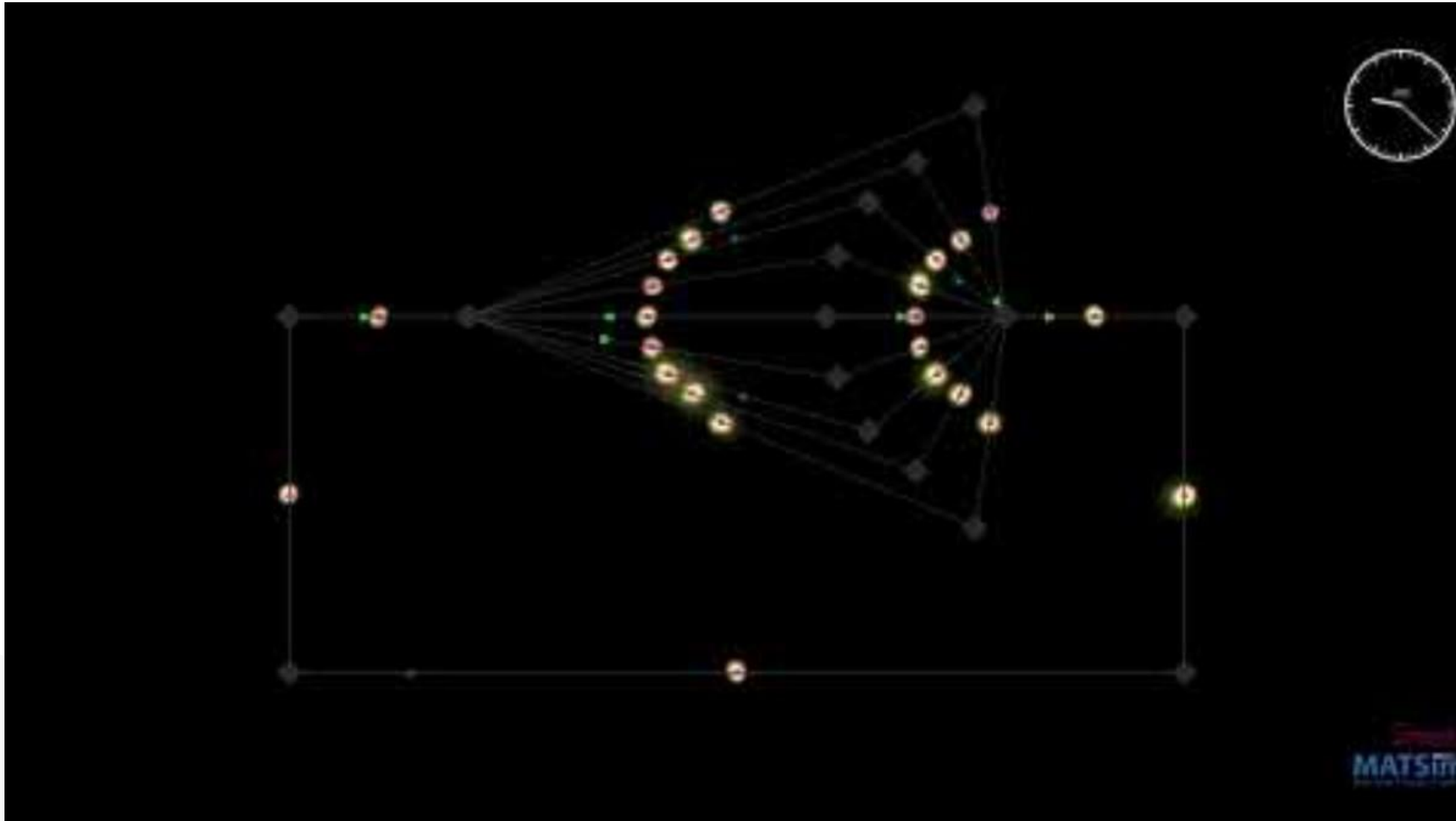
MATSIM uses evolutionary algorithm:

relies on random plan generation → requires iterations for optimization

Customizations – review

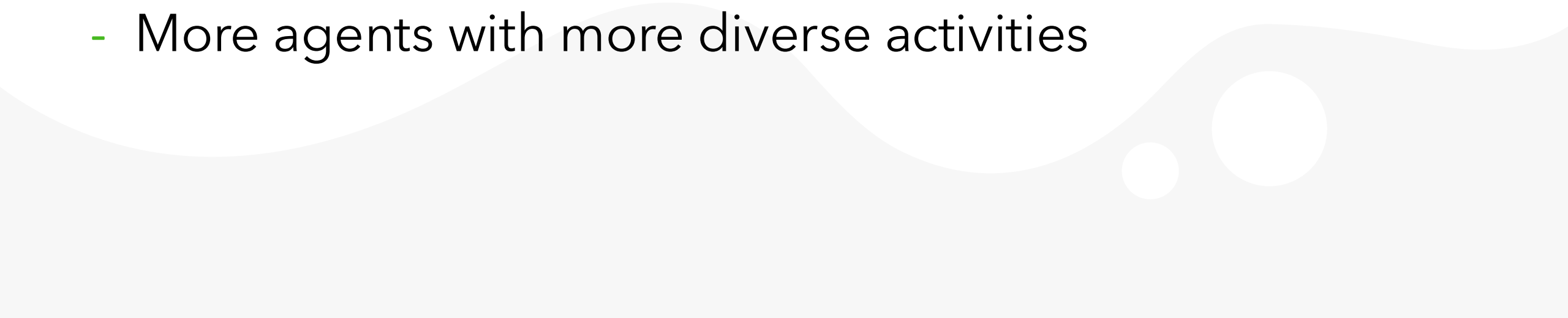
Scenario	Length	Capacity	Freespeed (m/s)	Agent	Activities
"Equil"	35000, 10000, 5000, 2000	36000, 3600, 1000, 360	27.78	100	Home, work
Scenario 1	10000, 5000	2500, 250	14.88	500	Home, work, school, other
Scenario 2	1500, 750	200, 100	14.88	500	Home, work, school, other
Scenario 3	1500, 750	80, 40	14.88	500	Home, work, school, other

Simulation



- Home
- Work
- School
- Other
- ▲ Facility
- ◆ Node
- Car

Comparisons to "equil" scenario - general

- Smaller in length and capacity
 - Lower freespeed
 - Bidirectional roads
 - More agents with more diverse activities
- 

Results – Average travel speeds

Scenario	Freespeed	Average travel distance	Average trip duration	Average travel speed
Baseline (equil)	27.78	44776.119	27:08	27.503
"equil", freespeed changed to 14.88	14.88	44776.119	50:28	14.787
"equil", with freespeed 14.88, 500 customized agents	14.88	65160	01:13:03	14.867
Scenario 1	14.88	19975	22:25	14.851
Scenario 2	14.88	3022.523	03:26	14.672
Scenario 3	14.88	3144	03:44	14.036

Comparisons – Conclusions

- As we decrease the capacity of the roads, the travel speed decreases
- The travel length affects the average travel distance
- The travel speed is dependent on the distance, capacity, and the freespeed
- Biggest factor of the travel speed is the freespeed, then the capacity

Results and comparisons – Iteration and average travel speed

- Remain relatively constant across iterations 0 to 50

Reason

- Simple network configuration
 - Fixed activity and time windows of agent behavior
- creates a relatively stable traffic flow pattern from early iterations (around 20)

Iteration	Average travel distance	Average travel time
0	3144	03:46
10	3144	03:45
20	3144	03:44
30	3144	03:44
40	3144	03:44
50	3144	03:44

Results and comparisons - Scores

- Scores

Scenario	20 iterations	50 iterations
Equil	67.17	67.4
Scenario 1	103.4	104.1
Scenario 2	110.0	110.0
Scenario 3	110.0	110.0

Conclusions

- Mainly depends on freespeed and road capacity
- Depends little on iteration

Conclusion— Key findings

Effect of Freespeed

- Most critical factor of travel speed
- 27.78 m/s to 14.88 m/s resulted in a significant  in travel speed from 27.503 m/s to 14.787 m/s.

Network Capacity Impact

- Lower capacities resulted in increased congestion and longer travel times
- Higher capacities showed improved traffic flow but may not reflect reality

Conclusion— Key findings

Agent Activity Patterns

- Create realistic urban mobility scenarios
- Incorporate diverse agent activity patterns
- Understand traffic flow & peak hours dynamics

Iteration and Optimization

- Scores of daily plans across 20 and 50 iterations remained relatively stable
- 20 iterations were sufficient for achieving optimized plans

Limitations

Simplicity of network

- 500 agents is too small to represent ~12 million trips (Transport Department)
- May not match the survey in every detail
- Our model excludes NHB and EB for simplicity
- Road lengths and capacities are hard to generalize

Limitations

Modifying Parameters in Scenarios

- Road lengths and capacities are hard to generalize
- Road characteristics vary significantly across Hong Kong (highways, local streets, arterials)
- Difficult to standardize the real-world road conditions

Our response to the limitations

Simplicity of network

- Our MATSim scenario is made simpler
- NHB and EB is more complex to implement
- Our 500-agent model can balance simplicity and data integrity

Our response to the limitations

Modifying Parameters in Scenarios

- Create 3 scenarios with varying lengths and capacities
- Scale down lengths and capacities → simulation size (500-agent model)
- Focused on relative differences rather than absolute accuracy

Conclusion

- Adjusting agent activities improved the realism of the simulation
 - Limitations: small sample of agents, simplicity of the "equil" network, uncertainty in travel behavior assumptions
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Thank **you!**

Any questions?

Q & A session